

Image 1 (main system diagram):

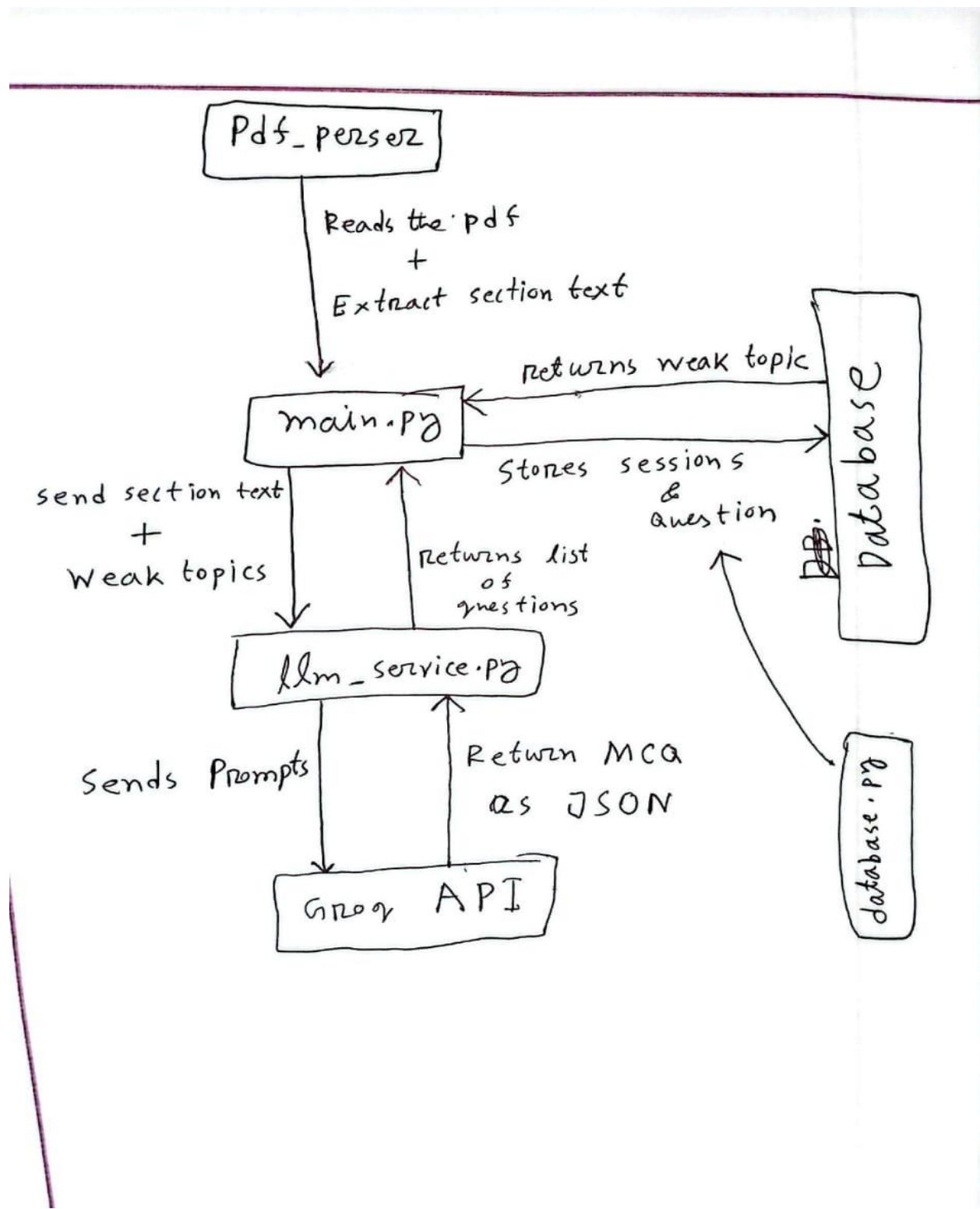
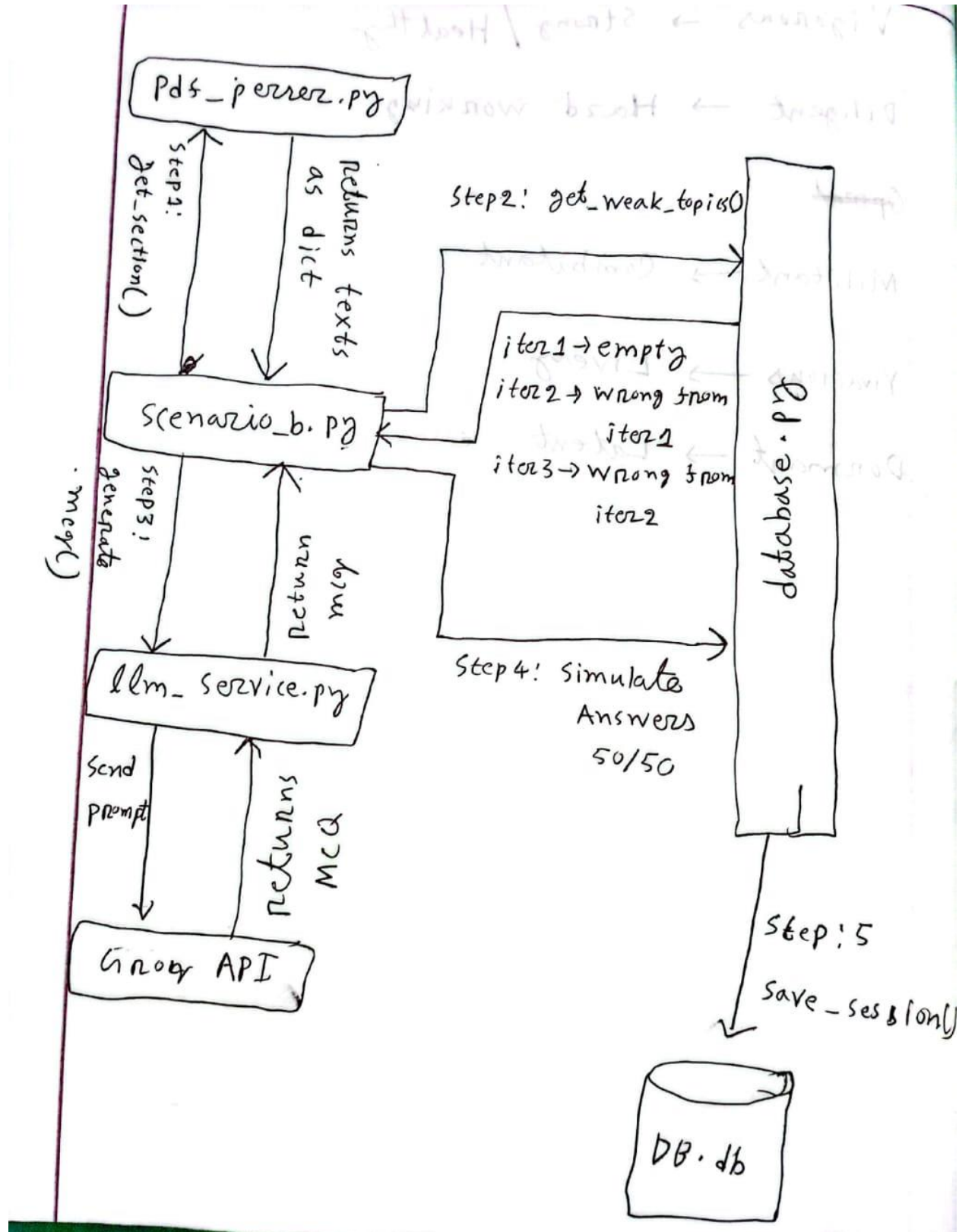


Image 1 (scenario_b diagram):



FILES AND WHAT THEY DO

pdf_perser.py

Reads the PDF and extracts section text.

get_all_sections(): returns all 10 sections as a dictionary

get_section_by_id(id): returns one section by its number

get_section_list() : returns section titles and short previews

database.py

Creates and manages the SQLite database.

init_db() : creates tables if they dont exist

save_session(ids, questions): saves session and all questions to db

get_weak_topics(ids) : returns most frequently wrong questions

get_kb_snapshot() : returns last 5 sessions

get_session_questions(id) : returns all questions from one session

llm_service.py

Sends section text to Groq API and gets MCQs back.

generate_mcqs(section_id, text, weak_topics, n):

returns list of questions each with choices, answer, explanation

main.py

FastAPI server that connects all files together.

GET /sections: returns list of all PDF sections

GET /sections/{id} : returns content of one section

POST /start-session : generates MCQs for selected sections

POST /submit-answers : scores answers and saves to database

GET /history : returns last 5 sessions

GET /history/{id} : returns all questions from one session

GET /weak-topics: returns most failed questions

scenario_b.py

Runs 3 automatic test iterations and saves output files.

Iteration 1: sections 5, 8 : no history, fresh start

Iteration 2: sections 6,8,9 : adapts from iteration 1

Iteration 3: section 8 : adapts from iteration 1 and 2